untangle::actuator

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1 actuator 1

2 Topic Index 5
2.1 Topics . . . 5

3 Namespace Index 7
3.1 Namespace List . . . 7

4 Concept Index 9
4.1 Concepts . . . 9

5 Class Index 11
5.1 Class List . . . 11

6 File Index 13
6.1 File List . . . 13

7 Topic Documentation 15
7.1 namespace untangle: functions . . . 15
7.1.1 Detailed Description . . . 16
7.1.2 Function Documentation . . . 16
7.1.2.1 connect() [1/4] . . . 16
7.1.2.2 connect() [2/4] . . . 17
7.1.2.3 connect() [3/4] . . . 18
7.1.2.4 connect() [4/4] . . . 19
7.1.2.5 bind() [1/3] . . . 19
7.1.2.6 bind() [2/3] . . . 20
7.1.2.7 bind() [3/3] . . . 20

8 Namespace Documentation 23
8.1 untangle Namespace Reference . . . 23
8.1.1 Detailed Description . . . 24
8.1.2 Typedef Documentation . . . 24
8.1.2.1 task_t . . . 24
8.1.3 Function Documentation . . . 25
8.1.3.1 last_arg() . . . 25
8.1.3.2 connect_one() . . . 25
8.1.3.3 connect_named_one() . . . 26
8.1.3.4 bind_task() . . . 27

9 Concept Documentation 29
9.1 untangle::task_callback_for Concept Reference . . . 29
9.1.1 Concept definition . . . 29
9.1.2 Detailed Description . . . 29
9.2 untangle::testable_for_emptiness Concept Reference . . . 30
9.2.1 Concept definition . . . 30

9.2.2 Detailed Description 30

10 Class Documentation 31
10.1 untangle::actuator< action_t > Struct Template Reference 31
10.1.1 Detailed Description 33
10.1.2 Member Typedef Documentation 34
10.1.2.1 actions_t 34
10.1.2.2 tasks_t 35
10.1.2.3 results_t 35
10.1.2.4 errors_t 35
10.1.3 Constructor & Destructor Documentation 35
10.1.3.1 actuator() 35
10.1.4 Member Function Documentation 36
10.1.4.1 operator=() 36
10.1.4.2 type() 36
10.1.4.3 reset() 36
10.1.4.4 copy_from() 37
10.1.4.5 release_owned() 37
10.1.4.6 invoke_callback() 38
10.1.4.7 take_callback() 39
10.1.4.8 operator()() 39
10.1.4.9 invoke_action() 40
10.1.4.10 call_tasks() 40
10.1.4.11 add() [1/4] 40
10.1.4.12 add() [2/4] 41
10.1.4.13 add() [3/4] 41
10.1.4.14 add() [4/4] 42
10.1.4.15 add_task() [1/2] 42
10.1.4.16 add_task() [2/2] 43
10.1.4.17 remove() [1/2] 43
10.1.4.18 remove() [2/2] 44
10.1.4.19 is_connected() 45
10.1.4.20 has_tasks() 45
10.1.4.21 has_action() 45
10.1.5 Member Data Documentation 46
10.1.5.1 tasks 46
10.1.5.2 owned 46
10.1.5.3 dispatching 46
10.2 untangle::invalid_action Struct Reference 47
10.2.1 Detailed Description 47
10.2.2 Constructor & Destructor Documentation 47
10.2.2.1 invalid_action() 47
10.2.3 Member Function Documentation 47

10.2.3.1 what() 47
10.3 untangle::actuator< action_t >::no_callback Struct Reference 48
10.3.1 Detailed Description 48

11 File Documentation 49
11.1 actuator.hpp 49

Index 57

Chapter 1

actuator

An actuator is a generic callable that may invoke a list of actions(callables of type std::function<...>).

Its goal is to provide a light and simple mechanism to connect components that want to communicate with each other. It works similar as the ”signal and slots” mechanisms from other frameworks, but much simplified.

Modern programming languages do not offer class intrinsic interfaces that would improve significantly how objects are connected and how they communicate. Instead, languages, like C++, are using external interfaces, that are not only cluttering the code, but they don't feel ”natural” as they are opposed to what an interface is in reality. Basically the objects and interfaces are not separated, and the interface is intrinsic to the object(think on the cogs in a mechanism). A generic callable is an attempt to mitigate this problem. async implementation is such use case.

An actuator provides the results of all actions in a vector, in the same order as the actions were added.

Example: how to use actuator instead of polymorphism

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle)
{
  for (const auto& s : shapes)
  {
    s->rotate(angle);
  }
}

std::shared_ptr<triangle> t(new triangle);
std::shared_ptr<circle> c(new circle);
std::shared_ptr<square> s(new square);

// using polymorphism
std::vector<shape*> shapes;
shapes.push_back(t.get());
shapes.push_back(c.get());
shapes.push_back(s.get());

rotate_shapes(shapes, 10);

// using actuator
auto action1 = untangle::bind(t, &triangle::rotate);
auto action2 = untangle::bind(c, &circle::rotate);
auto action3 = untangle::bind(s, &square::rotate);
auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

Anonymous lambdas as actions

An actuator normally does not own its actions: it stores pointers to std::function objects the caller keeps alive. add() also has an overload that takes an action by value, so an anonymous lambda can be an action with no named variable to keep around, and returns a handle to pass to remove():

```
untangle::actuator<std::function<int(int)» actuator_scale;
auto* handle = actuator_scale.add([](int v) { return v * 2; });
actuator_scale.add("triple", [](int v) { return v * 3; });
```

connect() accepts them too, but it deduces the action type from its arguments, and a lambda has its own closure type – it is not a std::function until something converts it. A call made only of anonymous lambdas therefore has nothing to deduce from, and the signature has to be named on connect() itself:

```
auto actuator_rotate = untangle::connect<std::function<void(int)»(
    [](int angle) { /* ... */ }, [](int angle) { /* ... */ });
```

Naming the type of the variable the result is assigned to does not supply it – template arguments are deduced from the call arguments alone, never from what the returned value is assigned to:

```
// ill formed: "no matching function for call to connect"
untangle::actuator<std::function<void(int)» actuator_rotate =
    untangle::connect([](int angle) { /* ... */ });
```

One named action anywhere in the call deduces the type for the whole of it, and the anonymous lambdas beside it then need nothing:

```
auto actuator_rotate = untangle::connect(action1, [](int angle) { /* ... */ });
```

The named form of connect() works the same way: a pair holding a pointer names an action the caller owns, a pair holding the action by value hands it over to the actuator, and the two can be mixed. The action type is deduced from the pointer, so a call whose first pair holds a lambda has to name the signature:

```
auto actuator_rotate = untangle::connect<std::function<void(int)»(
    std::make_pair("triangle", [](int angle) { /* ... */ }),
    std::make_pair("circle", [](int angle) { /* ... */ }));

// deduced from the leading pointer, so the lambda beside it needs nothing
auto actuator_mixed = untangle::connect(std::make_pair("triangle", &action1),
                                        std::make_pair("circle", [](int angle) { /* ... */ }));
```

Tasks: an action bound to its arguments and to the callback it must notify

An action is a subscription: it lives across invocations, and it is fired with a pack the caller supplies each time. A task is a one-shot. It carries its own arguments, is fired once, and reports what it produced to a callback it was built with. The two live side by side in one actuator and share nothing else – operator() fires the actions and leaves the tasks alone, call_tasks() fires the tasks and leaves the actions alone.

A task exists so that work which runs later than it was described can still say what became of it. That is what a queue needs, and an actuator holding only actions cannot give: the callback convention on operator() reads the trailing argument of the invocation, and a queued call has no invocation left to read from.

```
untangle::actuator<std::function<int(int)» actuator_scale;

// the callback comes last, after the arguments the action is bound to
actuator_scale.add_task(untangle::bind_task(action, 21, [](int result) { /* result == 42 */ }));

actuator_scale.call_tasks();  // runs it, notifies it, and empties the list
```

A task without a callback does not exist. That is the whole difference from an action's callback, which is optional and is recognised by its type – a trailing argument that does not look like a callback is simply an argument, and nothing fires. A task's callback is the last argument of bind_task(), taken by position, and task_callback_for only checks that it can serve. There is no answer "no" to fall through to, so a missing or unusable callback is a static_assert rather than silence.

It is not forwarded to the action either, so an action needs no callback parameter of its own – and a task whose action returns void is a task like any other. It reports with a void() callback, because finished is the message and the result is optional:

```
untangle::actuator<std::function<void(int)» actuator_notify;
actuator_notify.add_task(untangle::bind_task(action, 5, [] { /* finished */ }));
```

The arguments are copied when the task is built, while the caller still holds them, and handed to the action as the task's own lvalues when it runs. A task outlives the call that described it and cannot borrow what may already be gone; a move-only argument therefore does not compile.

add_task() answers bool. It refuses a task that has nothing to run or no callback to notify, and the refusal is the whole report – nothing is thrown and nothing is stored. Both are the caller's own mistake, and both are caught while the caller is still on the stack rather than surfacing later from call_tasks() as a std::bad_function_call recorded against a task whose action had in fact succeeded.

Finished does not mean failed. A task whose action throws is not notified: there is no result to hand over, and for a void task no completion to report either. What it threw goes to actuator::errors, the tasks behind it still run, and a caller relying on the callback alone will never hear about it. A callback runs inside the same try as the task that owns it, so what it throws travels the same path – errors can hold a failure for a task that succeeded.

A task's result goes to its callback and nowhere else: actuator::results is how an action hands back what it returned, and a task was built with something better. call_tasks() appends to errors rather than clearing it, while operator() clears both as it starts – so an actuator fired as one(); one.call_tasks(); reports both kinds together, and the actions go first.

Ask has_tasks() whether a task is waiting. is_connected() answers for the actions and for nothing else: an actuator holding only tasks is not connected, and a queue reading the wrong one would report a batch of tasks as nothing to do.

For convenience there are provided helpers methods to ”connect” to an initial list of ”actions”, or to create bindings to class methods.

Please check the manual in doc/refman.pdf for further references.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions 15

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle
Interface to untangle::actuator functor 23

Chapter 4

Concept Index

4.1 Concepts

Here is a list of all documented concepts with brief descriptions:

untangle::task_callback_for
: Can callback_t serve as the completion callback of a task returning result_t? 29

untangle::testable_for_emptiness
: Can callable_t say whether it is empty? 30

Chapter 5

Class Index

5.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

untangle::actuator< action_t >
: An actuator is a functor that can trigger a dynamic list of actions (of type std::function<...>) . . . 31
untangle::invalid_action
: Invalid action exception . . . 47
untangle::actuator< action_t >::no_callback . . . 48

Chapter 6

File Index

6.1 File List

Here is a list of all documented files with brief descriptions:

actuator.hpp 49

Chapter 7

Topic Documentation

7.1 namespace untangle: functions

Functions

- template<typename action_t, typename... Actions>
  actuator< action_t > untangle::connect (action_t &A1, Actions &&... An)

  Creates an actuator holding an initial list of actions.
- template<typename action_t, typename... Actions>
  actuator< action_t > untangle::connect (std::type_identity_t< action_t > &&A1, Actions &&... An)

  Creates an actuator owning an initial list of actions.
- template<typename action_t, typename key_t, typename... Actions>
  actuator< action_t > untangle::connect (std::pair< key_t, action_t * > A1, Actions &&... An)

  Creates an actuator holding an initial map of named actions.
- template<typename action_t, typename key_t, typename value_t, typename... Actions>
  requires (!std::is_same_v<value_t, action_t*>)
  actuator< action_t > untangle::connect (std::pair< key_t, value_t > A1, Actions &&... An)

  Creates an actuator owning an initial map of named actions.
- template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↩ T>::type>>
  action_t untangle::bind (const std::shared_ptr< class_t > &obj, T class_t::*method)

  Binding to a class function member.
- template<typename class_t, typename R, typename... Args>
  std::function< R(Args...)> untangle::bind (const std::shared_ptr< class_t > &obj, std::function< R(Args...)> class_t::*field)

  Binding to a funcref field: a std::function data member, called like a method.
- template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↩ T>::type>>
  action_t untangle::bind (class_t *obj, T class_t::*method)

  Binding to a class method.

7.1.1 Detailed Description

7.1.2 Function Documentation

7.1.2.1 connect() [1/4]

```
template<typename action_t, typename... Actions>
actuator< action_t > untangle::connect (
                action_t & A1,
                Actions &&... An)
```

Creates an actuator holding an initial list of actions.

Remarks

Ownership: an action passed as an lvalue is one the caller owns, and the actuator only points at it – it must outlive the actuator. An action passed as an rvalue, an anonymous lambda above all, is moved into the actuator instead and needs no named variable at all. The two can be mixed in one call. See the ownership convention on actuator.

Warning

A call made only of anonymous lambdas can not deduce action_t. A lambda has its own closure type and is not a std::function until something converts it, so there is nothing in such a call to deduce the action signature from, and it has to be named:

```
auto actuator_rotate = untangle::connect<std::function<void(int)»(
   [](int angle) { ... }, [](int angle) { ... });
```

Warning

Naming the type of the variable the result is assigned to does not supply it:

```
// still ill formed -- "no matching function for call to connect"
untangle::actuator<std::function<void(int)» actuator_rotate =
   untangle::connect([](int angle) { ... });
```

Template arguments are deduced from the call arguments alone, never from what the returned value is assigned to, so the explicit argument belongs on connect() itself. One named action anywhere in the call deduces action_t for the whole of it, and the anonymous lambdas beside it then need nothing.

Parameters

A1	- The first action. It is specified as std::function<...>.
An	- Any number of further actions, each convertible to the type of A1.

Returns

An actuator.

Example:

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle) {
 for (const auto& s : shapes) {
  s->rotate(angle);
 }
}
```

```
 auto action1 = untangle::bind(t, &triangle_mock::rotate);
 auto action2 = untangle::bind(c, &circle_mock::rotate);
 auto action3 = untangle::bind(s, &square_mock::rotate);

 auto actuator_rotate = untangle::connect(action1, action2, action3);
 actuator_rotate(20);

 // connect() deduces the action type from its arguments. A lambda has its own closure type
 // and is not a std::function until something converts it, so a call made only of anonymous
 // lambdas has nothing to deduce from and the signature has to be named on connect()
 // itself. Naming the type of the variable assigned to would not supply it: template
 // arguments are never deduced from what the returned value is assigned to.
 auto actuator_scale = untangle::connect<std::function<int(int)»([](int v) { return v * 2; },
                                                 [](int v) { return v * 3; });

 actuator_scale(10);
 ASSERT_THAT(actuator_scale.results, ::testing::ElementsAre(20, 30));
```

7.1.2.2 connect() [2/4]

```
template<typename action_t, typename... Actions>
actuator< action_t > untangle::connect (
                std::type_identity_t< action_t > && A1,
                Actions &&... An)
```

Creates an actuator owning an initial list of actions.

This overload takes the first action by rvalue, so it is the one that accepts a leading anonymous lambda. std::type_identity_t makes action_t a non-deduced context here, which is what both keeps this overload out of the way of the lvalue one – it is not a candidate at all unless action_t is named – and states the requirement described by the warning on the overload taking the first action by lvalue.

Parameters

A1	- The first action, as an rvalue.
An	- Any number of further actions, owned or pointed at by value category.

Returns

An actuator.

7.1.2.3 connect() [3/4]

```
template<typename action_t, typename key_t, typename... Actions>
actuator< action_t > untangle::connect (
                std::pair< key_t, action_t * > A1,
                Actions &&... An)
```

Creates an actuator holding an initial map of named actions.

Remarks

Ownership works as it does for the unnamed connect() overloads: a pair holding a pointer names an action the caller owns, and a pair holding an action by value – an anonymous lambda above all – hands it to the actuator to own. The two can be mixed in one call.

Warning

The action type is deduced from the pointer in the first pair, so a call whose first pair holds an anonymous lambda has nothing to deduce it from and has to name the signature on connect() itself, exactly as the unnamed overload does:

```
auto actuator_rotate = untangle::connect<std::function<void(int)»(
    std::make_pair("triangle", [](int angle) { ... }),
    std::make_pair("circle", [](int angle) { ... }));
```

Remarks

action_t leads the template parameter list, ahead of key_t, so that naming it – connect<std::↩ function<...>>(...) – means the same thing on every connect() overload. Both parameters are deduced from the first pair whenever it holds a pointer, so nothing has to be named here in the first place.

Parameters

A1	- The first name/action pair. Its second element is a pointer to an action.
An	- Any number of further name/action pairs, owned or pointed at according to whether they hold the action itself or a pointer to it.

Returns

An actuator.

Example:

```
// A pair holding the action itself hands it to the actuator to own. The action type is
// deduced from a pointer, and there is no pointer here, so the signature has to be named
// on connect() -- as it does for the unnamed overload.
auto actuator_scale = untangle::connect<std::function<int(int)»(
    std::make_pair("double", [](int v) { return v * 2; }),
    std::make_pair("triple", [](int v) { return v * 3; }));

actuator_scale.invoke_action("double", 10);
ASSERT_THAT(actuator_scale.results, ::testing::ElementsAre(20));
actuator_scale.invoke_action("triple", 10);
ASSERT_THAT(actuator_scale.results, ::testing::ElementsAre(30));
```

7.1.2.4 connect() [4/4]

```
template<typename action_t, typename key_t, typename value_t, typename... Actions>
requires (!std::is_same_v<value_t, action_t*>)
actuator< action_t > untangle::connect (
                std::pair< key_t, value_t > A1,
                Actions &&... An)
```

Creates an actuator owning an initial map of named actions.

This overload takes a first pair holding the action itself rather than a pointer to it, so it is the one that accepts a leading anonymous lambda. action_t is never deduced here – the requires clause is what keeps it from competing with the overload taking a pointer in the first pair over a leading pointer pair, which both would otherwise accept once action_t is named.

Parameters

A1	- The first name/action pair, holding the action itself.
An	- Any number of further name/action pairs.

Returns

An actuator.

7.1.2.5 bind() [1/3]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::↩
type>>
action_t untangle::bind (
                const std::shared_ptr< class_t > & obj,
                T class_t::* method)
```

Binding to a class function member.

It returns a std::function(lambda) that wraps the function member. It may be used to provide an action for connect() or actuator::add().

Remarks

It requires a shared pointer to the class type. A std::weak_ptr to it is captured internally in a lambda, so the binding can always check whether the shared object is still alive without keeping it alive itself. Therefore, it is safe to use actions provided by this binding inside an actuator, and safe for the action to outlive the caller's shared pointer.

Parameters

obj	- Class object.

method	- Pointer to function member. It is specified as &<class type>::<function member>

Returns

action_t - A std::function that wraps the pointer to function member.

Remarks

If the class object gets invalid, invoking this binding will throw an exception of type invalid_action.

7.1.2.6 bind() [2/3]

```
template<typename class_t, typename R, typename... Args>
std::function< R(Args...)> untangle::bind (
                const std::shared_ptr< class_t > & obj,
                std::function< R(Args...)> class_t::* field)
```

Binding to a funcref field: a std::function data member, called like a method.

It holds the class object as the method overload does, through a std::weak_ptr, and reads the field on every call, so a later assignment to the field is what runs.

Parameters

obj	- Class object.
field	- Pointer to the std::function data member, as &<class type>::<field>.

Returns

A std::function that calls the field.

Remarks

Invoking this binding throws invalid_action when the class object is gone or the field is empty, so an actuator drops it either way.

7.1.2.7 bind() [3/3]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::↩
type>>
action_t untangle::bind (
                class_t * obj,
                T class_t::* method)
```

~~Binding to a class method.~~

Attention

It is not safe to use this binding when the pointed-to object may be destroyed. A null pointer is

Parameters

obj	- Pointer to class.
method	- Pointer to function member. It is specified as &<class type>::<function member>

Returns

action_t - A std::function that wraps the pointer to function member.

Chapter 8

Namespace Documentation

8.1 untangle Namespace Reference

Interface to untangle::actuator functor.

Classes

- struct invalid_action
  Invalid action exception.
- struct actuator
  An actuator is a functor that can trigger a dynamic list of actions (of type std::function<...>).

Concepts

- concept task_callback_for
  Can callback_t serve as the completion callback of a task returning result_t?
- concept testable_for_emptiness
  Can callable_t say whether it is empty?

Typedefs

- using task_t = std::function<bool()>
  An action, its arguments and the callback it notifies, sealed into one nullary callable.

Functions

- template<typename... Args>
  auto last_arg (Args &&... args)

  Get last argument of params pack;.
- template<typename action_t, typename arg_t>
  void connect_one (actuator< action_t > &target, arg_t &&arg)

  Add one connect() argument to the actuator being built.
- template<typename action_t, typename... Actions>
  actuator< action_t > connect (action_t &A1, Actions &&... An)

  Creates an actuator holding an initial list of actions.
- template<typename action_t, typename... Actions>
  actuator< action_t > connect (std::type_identity_t< action_t > &&A1, Actions &&... An)

  Creates an actuator owning an initial list of actions.
- template<typename action_t, typename entry_t>
  void connect_named_one (actuator< action_t > &target, entry_t &&entry)

  Add one named connect() argument to the actuator being built.
- template<typename action_t, typename key_t, typename... Actions>
  actuator< action_t > connect (std::pair< key_t, action_t * > A1, Actions &&... An)

  Creates an actuator holding an initial map of named actions.
- template<typename action_t, typename key_t, typename value_t, typename... Actions>
  requires (!std::is_same_v<value_t, action_t*>)
  actuator< action_t > connect (std::pair< key_t, value_t > A1, Actions &&... An)

  Creates an actuator owning an initial map of named actions.
- template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↩ T>::type>>
  action_t bind (const std::shared_ptr< class_t > &obj, T class_t::*method)

  Binding to a class function member.
- template<typename class_t, typename R, typename... Args>
  std::function< R(Args...)> bind (const std::shared_ptr< class_t > &obj, std::function< R(Args...)> class_t::*field)

  Binding to a funcref field: a std::function data member, called like a method.
- template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↩ T>::type>>
  action_t bind (class_t *obj, T class_t::*method)

  Binding to a class method.
- template<typename action_t, typename... Args>
  task_t bind_task (action_t &&action, Args &&... args)

  Binds an action to its arguments and to the callback it must notify, which comes last.

8.1.1 Detailed Description

Interface to untangle::actuator functor.

8.1.2 Typedef Documentation

8.1.2.1 task_t

using untangle::task_t = std::function<bool()>

An action, its arguments and the callback it notifies, sealed into one nullary callable.

What bind_task() builds, actuator::add_task() holds and actuator::call_tasks() fires.

Remarks

An action is a subscription; a task is a one-shot: it carries its own arguments, is fired once and reports to its own callback – so it is held by value, and call_tasks() consumes it.

Running it is what reports: the result goes to the callback and nowhere else. What it returns says only whether it did its job - true once a task from bind_task() has run and notified - and a caller may ignore it. A throw is not turned into false: it propagates.

It names neither the argument types nor the result type, so one container holds tasks built from completely different actions. Anything that can be a std::function<bool()> is a task.

8.1.3 Function Documentation

8.1.3.1 last_arg()

```
template<typename... Args>
auto untangle::last_arg (
                Args &&... args)
```

Get last argument of params pack;.

Template Parameters

Args	params template types

Parameters

args	params

Returns

last param value in case it exist, otherwise 0.

8.1.3.2 connect_one()

```
template<typename action_t, typename arg_t>
void untangle::connect_one (
                actuator< action_t > & target,
                arg_t && arg)
```

Add one connect() argument to the actuator being built.

An argument that is an lvalue of the action type is one the caller owns: only its address is stored, and it has to outlive the actuator. Anything else – an anonymous lambda, a temporary std::function, a named lambda that is not an action_t yet – has no owner to outlive the actuator, so it is converted to an action and moved into it.

Remarks

Empty actions are dropped rather than stored, which is what connect() has always done with them. Dropping an empty owned action before it is stored is what keeps actuator::owned free of entries

Template Parameters

action_t	Action type of the actuator being built.
arg_t	Deduced type of the argument, carrying its value category.

Parameters

target	- The actuator being built.
arg	- One connect() argument.

8.1.3.3 connect_named_one()

```
template<typename action_t, typename entry_t>
void untangle::connect_named_one (
                actuator< action_t > & target,
                entry_t && entry)
```

Add one named connect() argument to the actuator being built.

It is connect_one() for the named overloads: the second element of the pair decides ownership. A pointer to an action is one the caller owns and is only stored; anything else – an anonymous lambda, a temporary std::function – is converted to an action and moved into the actuator.

Remarks

A null pointer is dropped rather than stored, which is what the named connect() has always done with one. An owned action that is empty is dropped for the same reason the unnamed connect_one() drops one: nothing would ever point at it.

Template Parameters

action_t	Action type of the actuator being built.
entry_t	Deduced type of the pair, carrying its value category.

Parameters

target	- The actuator being built.
entry	- One name/action pair.

8.1.3.4 bind_task()

```
template<typename action_t, typename... Args>
task_t untangle::bind_task (
                action_t && action,
                Args &&... args)
```

Binds an action to its arguments and to the callback it must notify, which comes last.

Running the task_t it returns invokes the action and then hands the callback what it returned. The arguments are copied here, while the caller still holds them, and handed to the action as the task's own lvalues – a task runs later than it was built.

Remarks

The callback is the last argument, by position, not recognised by its type as an action's is: a pack cannot be followed by a deducible parameter, so it is split off here and callers above just forward (action, args...). task_callback_for only asks whether it can serve, which is what lets it be required. It is not forwarded to the action, which therefore needs no callback parameter of its own – and a void action is a task like any other.

Attention

Each argument is copied, so a move-only one does not compile. The result is not: it is moved into the callback, which may take it by rvalue reference – see task_callback_for.

Remarks

The action goes into the task once: copied from an lvalue, moved from an rvalue, and not moved again on the way. A queue builds a task per submission, so a by-value parameter here would be a move more for every one.

Attention

An empty callback or an empty action builds an empty task, which actuator::add_task() refuses. Only a callable that can be tested for emptiness is tested, so a bare lambda is never refused.

Template Parameters

action↩ _t	Type of the action as passed: a reference for an lvalue, so its decayed type is the one stored, and that may be any callable naming a result_type.
Args	Types of the arguments to bind, of which the last is the callback.

Parameters

action	- The action to bind.
args	- The arguments to bind to action, followed by the callback to notify.

Returns

The task, or an empty one – see above.

Chapter 9

Concept Documentation

9.1 untangle::task_callback_for Concept Reference

Can callback_t serve as the completion callback of a task returning result_t?

#include <actuator.hpp>

9.1.1 Concept definition

```
template<typename callback_t, typename result_t>
concept task_callback_for = requires(callback_t& c, result_t&& r) {
  requires std::is_void_v<decltype(c(std::forward<result_t>(r)))>;
} || (std::is_void_v<result_t> && requires(callback_t& c) {
                                    requires std::is_void_v<decltype(c())>;
                                  })
```

9.1.2 Detailed Description

Can callback_t serve as the completion callback of a task returning result_t?

It asks only whether it can serve, never whether it was meant as one: bind_task() takes the callback by position. A void result reports finished, so its callback takes no argument.

Remarks

The result is handed over as an rvalue, because a task keeps nothing of it: a callback may take it by value, by const reference or by rvalue reference, and a move-only result is reported like any other. A callback taking a non-const lvalue reference cannot be handed it.

Attention

Both disjuncts are load-bearing: the first yields false for a void result rather than failing hard (void&& in a requires-parameter list is a substitution failure in the immediate context), and the second's std::is_void_v guard keeps a std::function<void()> from satisfying this for any result type.

Template Parameters

callback↩ _t	Type of the callback.
result_t	Result type of the task it reports on.

9.2 untangle::testable_for_emptiness Concept Reference

Can callable_t say whether it is empty?

#include <actuator.hpp>

9.2.1 Concept definition

```
template<typename callable_t>
concept testable_for_emptiness =
    std::is_pointer_v<callable_t> || requires(const callable_t& c) { c.operator bool(); }
```

9.2.2 Detailed Description

Can callable_t say whether it is empty?

bind_task() refuses an empty action or an empty callback, and has to decide first whether that question can be put at all. A std::function answers it with operator bool, and a pointer with its own value.

Attention

A lambda is not asked, and asking anyway is worse than not asking. A captureless one converts implicitly to a function pointer, so a plain static_cast<bool> test compiles and answers on its behalf – always true, because that conversion never yields null. The test is dead for every lambda that reaches it, and MSVC reports the conversion it rests on as a truncation (C4305). Asking for operator bool by name is what leaves a lambda out.

Template Parameters

callable↩ _t	Type of the action or the callback.

Chapter 10

Class Documentation

10.1 untangle::actuator< action_t > Struct Template Reference

An actuator is a functor that can trigger a dynamic list of actions (of type std::function<...>).

#include <actuator.hpp>

Classes

- struct no_callback

Public Types

- using actions_t = std::vector<action_t*>
  Actions container type.
- using tasks_t = std::deque<task_t>
  Tasks container type.
- using results_t = std::vector<typename result_t::type>
  Results container type.
- using errors_t = std::vector<std::exception_ptr>
  Errors container type.

Public Member Functions

- actuator (const actuator &other)
  Copy constructor.
- actuator & operator= (const actuator &other)
  Assignment operator.
- action_t type () const
  Helper method to access the action type of this object.
- bool reset ()
  Remove all actions and tasks, and any stored results and errors.
- void copy_from (const actuator &other)
  Copy another actuator, re-pointing the copied actions at this object's storage.
- void release_owned (const action_t *action)

Destroy an owned action that nothing points at any more.

- template<typename last_t>
  void invoke_callback (last_t &last)

  Invoke the trailing callback argument, if the invocation has one.
- template<typename... Args>
  void operator() (Args &&... args)

  The call operator.
- template<typename... Args>
  void invoke_action (const std::string &name, Args &&... args)

  Invokes one single action associated with a key.
- void call_tasks ()

  Fires every task held, in the order add_task() took them, and empties the list.
- void add (action_t *action)

  Add an action to the actions list.
- action_t * add (action_t &&action)

  Add an action the actuator owns.
- void add (const std::string &name, action_t *action)

  Add action to the actions map associated with a name.
- action_t * add (const std::string &name, action_t &&action)

  Add an action the actuator owns, associated with a name.
- bool add_task (task_t &&task)

  Adds a task, to be fired by call_tasks().
- template<typename callable_t>
  requires std::is_void_v<std::invoke_result_t<std::decay_t<callable_t>&>>
  bool add_task (callable_t &&callable)

  Adds a nullary callable that returns nothing, as a task that answers true once it has run.
- bool remove (const action_t *action)

  Remove an action from the actions list.
- bool remove (const std::string &name)

  Remove an action from actions map.
- bool is_connected () const

  Check if this actuator is "connected" with other actions.
- bool has_tasks () const

  Is there a task waiting to be fired?
- bool is_dispatching () const

  Whether an action is being called right now, so a removal would be refused.
- bool has_action (const std::string &name) const

  Check if there is certain named action.

Static Public Member Functions

- template<typename... Args>
  static auto take_callback (Args &... args)

  Copies the trailing callback out of args, or nothing when the last argument is none.

Public Attributes

- actions_t actions

  Actions list.
- actions_map_t actions_map

  Named actions map.
- results_t results

  Actions return values list.
- errors_t errors

  What the actions threw. See errors_t.
- tasks_t tasks

  Tasks waiting to be fired, in the order add_task() took them.
- std::list< action_t > owned

  Actions this actuator owns, as added by add(action_t&&).
- std::size_t dispatching = 0

10.1.1 Detailed Description

template<typename action_t>
struct untangle::actuator< action_t >

An actuator is a functor that can trigger a dynamic list of actions (of type std::function<...>).

Remarks

An actuator object can be constructed with an initial list of actions by connect().

Callback convention: when the last argument of an invocation can be called with the action return type R and returns nothing, it is not passed on as a plain argument alone – it is also treated as a completion callback. The action is invoked with the full argument list as usual, and the callback is then invoked with the action return value. Any void returning callable qualifies: a std::function<void(R)>, a lambda taking R, a function pointer. It applies to operator()() and to invoke_action() alike. Actions returning void have no result to report, so no callback is invoked for them.

Warning

The return type is what separates a callback from an argument the action means to consume itself. A trailing callable that returns a value – a transform, a comparator – is left alone, and by the same rule a callback written to return something is silently not invoked. A callback returns nothing.

Remarks

Argument convention: an action must not take ownership of the arguments it is invoked with. operator()() forwards one argument pack to every action in the list, and an rvalue can be moved from only once, so an action that consumes an argument leaves the actions after it holding a moved-from object – an empty std::function for a callback a later action means to call itself. The arguments are forwarded rather than copied so that a single action pays nothing for the broadcast; what that buys has to be respected by the actions.

Warning

An action does not have to move an argument in its body to consume it: a parameter taken by value is move constructed from an rvalue by the call itself. Take arguments by reference or const reference, or invoke with lvalues, whenever more than one action is connected. The behaviour is otherwise unpredictable, and it is the caller and the action signatures together that decide it – an actuator cannot detect the violation.

Remarks

invoke_action() invokes one single action, so nothing follows it and the convention does not constrain it.

Failure convention: an action that throws does not stop the invocation. What it threw is stored in actuator::errors, one std::exception_ptr per failure, and the actions after it still run. An untangle::invalid_action means a dead binding, so that action is dropped as well; anything else leaves the action in place. Nothing is printed and nothing escapes the call - the caller reads actuator::errors and decides. A failed action pushes no result, so actuator::results then holds fewer entries than there are actions.

Ownership convention: an actuator normally does not own its actions. It stores pointers to std::↩ function objects the caller keeps alive, and those have to outlive it. add(action_t&&) is the exception – it moves the action into actuator::owned and points at the stored copy – which is what lets an anonymous lambda be an action, with no named variable to keep around. The handle it returns is the only way to remove() such an action afterwards.

Re-entrancy convention: nothing is removed while the actuator is dispatching - while operator()() or invoke_action() is calling an action. Removal destroys an owned action at once, so the action running, or the loop calling it, would be left on freed memory. Both remove() overloads and reset() refuse it and return false; remove once the call has returned. Adding during a dispatch is fine: an added action is called in the same pass.

Template Parameters

action↩ _t	Action type. It is specified as std::function<...>.

10.1.2 Member Typedef Documentation

10.1.2.1 actions_t

template<typename action_t>
using untangle::actuator< action_t >::actions_t = std::vector<action_t*>

Actions container type.

Remarks

The elements stored are of pointer type, that is required to implement the remove() operation. std::function supports only equality operator for nullptr (two std::function(s) can not compare).

A std::vector, walked on every dispatch: contiguous pointers cost a receiver one load less than a std::list's nodes. operator()() goes by index, so an action added during a dispatch, which may reallocate, is still called in the same pass.

10.1.2.2 tasks_t

```
template<typename action_t>
using untangle::actuator< action_t >::tasks_t = std::deque<task_t>
```

Tasks container type.

Remarks

The elements are values, not pointers: identity is what remove() needs of an action, and no one removes a single task – call_tasks() consumes the whole list.

A std::deque, which stores its tasks in blocks: a queue adds one per submission, and a std::list paid an allocation for each. Nothing keeps a reference into it - call_tasks() takes the whole list before it fires a task. It names no result type, because untangle::task_t does not.

10.1.2.3 results_t

```
template<typename action_t>
using untangle::actuator< action_t >::results_t = std::vector<typename result_t::type>
```

Results container type.

It holds the return values of the actions that have a non-void return type. Upon the actuator invocation, the returns can be extracted from results.

Remarks

One entry per action that returned, in invocation order. An action that threw leaves none, so results[i] is the i-th action's return only when actuator::errors is empty.

10.1.2.4 errors_t

```
template<typename action_t>
using untangle::actuator< action_t >::errors_t = std::vector<std::exception_ptr>
```

Errors container type.

It holds what the actions threw, as std::exception_ptr, in the order they were invoked.

10.1.3 Constructor & Destructor Documentation

10.1.3.1 actuator()

```
template<typename action_t>
untangle::actuator< action_t >::actuator (
                const actuator< action_t > & other)   [inline]
```

Copy constructor.

Remarks

It can not be defaulted, see copy_from().

10.1.4 Member Function Documentation

10.1.4.1 operator=()

```
template<typename action_t>
actuator & untangle::actuator< action_t >::operator= (
                const actuator< action_t > & other)   [inline]
```

Assignment operator.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
untangle::actuator<decltype(actuator_rotate.type())> actuator_rotate_1 = actuator_rotate;
actuator_rotate_1(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

10.1.4.2 type()

```
template<typename action_t>
action_t untangle::actuator< action_t >::type () const   [inline]
```

Helper method to access the action type of this object.

Note

Intended to be used in expressions by decltype(<actuator instance>.type()

Returns

action_t

10.1.4.3 reset()

```
template<typename action_t>
bool untangle::actuator< action_t >::reset ()   [inline]
```

Remove all actions and tasks, and any stored results and errors.

After this call the actuator is empty: actuator::is_connected() and actuator::has_tasks() return false. The actions it owns are destroyed, so every handle returned by add(action_t&&) is dangling afterwards; the actions it merely points at are left untouched.

Returns

true - emptied. false - refused during a dispatch, and nothing changed: see the re-entrancy convention on actuator.

10.1.4.4 copy_from()

```
template<typename action_t>
void untangle::actuator< action_t >::copy_from (
                const actuator< action_t > & other)   [inline]
```

Copy another actuator, re-pointing the copied actions at this object's storage.

Remarks

A defaulted copy is wrong as soon as an actuator owns anything: it duplicates actuator::owned but leaves actuator::actions pointing into other's copy of it, so the new object dangles the moment other is destroyed. The owned actions are copied first, then each pointer into other::owned is translated to the matching element here.

A pointer to an action the caller owns is external to both actuators and is copied unchanged: it is not in either actuator::owned, so it is not in the translation table and is left as it is.

Parameters

other	- The actuator to copy. Self copy is the caller's to exclude.

10.1.4.5 release_owned()

```
template<typename action_t>
void untangle::actuator< action_t >::release_owned (
                const action_t * action)   [inline]
```

Destroy an owned action that nothing points at any more.

Remarks

Removing an action only drops a pointer to it. When the actuator owns that action the std::function itself lives in actuator::owned and has to go too, or a loop of add(action_t&&) and remove() grows actuator::owned without bound.

The action is kept while anything still refers to it: one handle can be added to both the list and the map, and removing it from one must not leave the other pointing at a destroyed object. An action the actuator does not own is not in actuator::owned, so nothing is found to erase and the caller's object is left alone.

Parameters

action	- The action just removed. A null pointer is accepted and matches nothing.

10.1.4.6 invoke_callback()

```
template<typename action_t>
template<typename last_t>
void untangle::actuator< action_t >::invoke_callback (
                last_t & last)  [inline]
```

Invoke the trailing callback argument, if the invocation has one.

Does nothing unless last_t can be called with the action return type and returns nothing, so any other trailing argument is left alone. The test is on what the type can do, not on what it is, so a plain lambda qualifies as well as a std::function. See the callback convention on actuator.

Remarks

The single condition does the work of two tests: (1) whether last can be called with the action return type at all, and (2) whether that call returns nothing. Spelled out as traits they need two nested if constexpr:

```
// (1)
if constexpr (std::is_invocable_v<last_t&, typename result_t::type>) {
  // (2)
  if constexpr (std::is_void_v<std::invoke_result_t<last_t&, typename result_t::type»>) {
    last(results.back());
  }
}
```

Both are folded into the one expression instead, each in a different part of it:

```
requires { requires std::is_void_v<decltype( last(results.back()) )>; }
//         \_____________ (2) _____________/          \________ (1) ________/
```

Remarks

(1) Invocability comes for free. To form decltype(last(results.back())) the compiler must first form the call last(...). If last is an int, that expression does not exist – and inside a requires-expression a substitution failure yields false instead of being ill formed, the same property concepts are built on. So invocability is never tested separately: it is a precondition of the return type question, and failing it is what makes the whole condition false.

(2) The return type test is the inner requires, and the doubled keyword is not a typo. requires { expr; } asks only whether expr is well formed, and std::is_void_v<...> is well formed when it is false too – that spelling would let a value returning callable through. The second keyword makes it a nested requirement: a condition that has to hold, not merely compile.

The trait spelling has to nest, which is part of why it is not the one used here: the two if constexpr cannot be joined with &&. && short circuits evaluation, not instantiation, so std::invoke_result_t is formed whatever the first trait says, and for a type that is not invocable at all it has no ::type – a hard error rather than a false result.

The caller must copy the callback out of the argument pack before invoking the action: an action taking it by value moves from the caller's std::function, which would leave this an empty function to call.

Template Parameters

last_t	Type of what take_callback() returned.

Parameters

last	- The callback, copied before the action was invoked, or a no_callback.

10.1.4.7 take_callback()

```
template<typename action_t>
template<typename... Args>
auto untangle::actuator< action_t >::take_callback (
                Args &... args)   [inline], [static]
```

Copies the trailing callback out of args, or nothing when the last argument is none.

Remarks

Only a callback is copied. An ordinary last argument is never read from here, so a copy of it would be pure cost – a whole payload per call for a caller passing a large one.

The test is invoke_callback()'s own: an action returning void reports nothing, so it has no callback whatever its last argument is.

It is not task_callback_for, which hands the result over as an rvalue. The result here stays in actuator::results and is handed over as an lvalue, so a callback taking it by rvalue reference is an ordinary argument: nothing is moved out of the results.

10.1.4.8 operator()()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::operator() (
                Args &&... args)   [inline]
```

The call operator.

Actions in the actuator::actions list are triggered by invoking the call operator.

Parameters

args	- Arguments list must match the action arity. A trailing callback is invoked once per action, with that action's return value; see the convention on actuator.

Warning

The same argument pack is forwarded to every action in the list, so the argument convention on actuator applies here in full: an action that consumes an argument leaves the actions after it with a moved-from object.

Remarks

Nothing escapes this call: what an action throws goes to actuator::errors, and the actions after it still run. See the failure convention on actuator.

10.1.4.9 invoke_action()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::invoke_action (
                const std::string & name,
                Args &&... args)  [inline]
```

Invokes one single action associated with a key.

An action that can not be invoked – a null pointer, an empty std::function, a binding whose object is gone – is dropped from actuator::actions_map instead, exactly as operator()() drops one from actuator::actions. Nothing is invoked and nothing escapes; actuator::has_action() reports it gone afterwards.

Remarks

What the action throws is stored in actuator::errors, as it is for operator()(). See the failure convention on actuator.

Parameters

name	- Key associated with the action. Invoking a key that is not in the map does nothing.
args	- Arguments list must match the action arity. A trailing callback is invoked with the action's return value; see the convention on actuator.

10.1.4.10 call_tasks()

```
template<typename action_t>
void untangle::actuator< action_t >::call_tasks ()  [inline]
```

Fires every task held, in the order add_task() took them, and empties the list.

Each task notifies its own callback as it runs, and its result goes there alone – never to actuator::results. It does not fire the actions, as operator()() does not fire the tasks.

Remarks

Errors are appended to actuator::errors, which operator()() clears instead – so one(); one.call_↩ tasks(); reports both kinds together, and the actions go first.

Attention

Finished does not mean failed. A task that throws does not stop the pass and is not notified; a callback that throws lands in actuator::errors the same way. The task is consumed either way, so a caller waiting on the callback alone must read the errors.

10.1.4.11 add() [1/4]


```
template<typename action_t>
void untangle::actuator< action_t >::add (
                action_t * action)  [inline]
```

Parameters

action	- Action to be added.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(2).WillRepeatedly(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.add(&action1);
actuator_rotate(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

10.1.4.12 add() [2/4]

```
template<typename action_t>
action_t * untangle::actuator< action_t >::add (
                action_t && action)   [inline]
```

Add an action the actuator owns.

The action is moved into actuator::owned, so it needs no named variable outliving the actuator: an anonymous lambda can be passed straight in, and the call converts it to action_t. See the ownership convention on actuator.

Parameters

action	- Action to be added. It is taken by value and moved from.

Returns

action_t* - A handle to the stored action, to pass to remove(). It stays valid until that action is removed or the actuator is destroyed. A copy of the actuator owns its own copy of the action and has its own handle to it.

Example:

```
// An anonymous lambda has no named variable to outlive the actuator, so the actuator takes
// ownership of it: it is moved into actuator::owned and the list points at the stored copy.
untangle::actuator<std::function<int(int)» actuator_scale;
actuator_scale.add([](int v) { return v * 2; });
actuator_scale.add([](int v) { return v * 3; });

actuator_scale(10);
ASSERT_THAT(actuator_scale.results, ::testing::ElementsAre(20, 30));
```

10.1.4.13 add() [3/4]


```
template<typename action_t>
void untangle::actuator< action_t >::add (
                const std::string & name,
```

Parameters

name	- Name of the action.
action	- Action to be added.

10.1.4.14 add() [4/4]

```
template<typename action_t>
action_t * untangle::actuator< action_t >::add (
                const std::string & name,
                action_t && action)   [inline]
```

Add an action the actuator owns, associated with a name.

Parameters

name	- Name of the action.
action	- Action to be added. It is taken by value and moved from.

Returns

action_t* - A handle to the stored action, as add(action_t&&) returns, or nullptr when name is already taken. The name already in the map keeps the action it has, as it does for add(const std::string&, action_t*), and the action offered here is dropped rather than left ownerless in actuator::owned.

10.1.4.15 add_task() [1/2]

```
template<typename action_t>
bool untangle::actuator< action_t >::add_task (
                task_t && task)   [inline]
```

Adds a task, to be fired by call_tasks().

It is stored by value: an actuator owns every task it holds, unlike the actions it merely points at. Anything that can be an untangle::task_t is taken, from bind_task() or not. There is no counterpart to remove() and no named form – call_tasks() consumes the list, and tasks fire as a batch, in the order they were added.

Attention

It refuses an empty task, and the refusal is the whole report: nothing is thrown and nothing is stored. It lands while the caller is still on the stack, rather than surfacing from call_tasks() as a std::bad_function_call read as the failure of a task that had in fact run.

Parameters

task	- The task to add. Taken by value and moved from; a refused one is dropped.

Returns

true - taken, and call_tasks() will fire it.

false - refused: it is empty, so it can neither run nor report.

10.1.4.16 add_task() [2/2]

```
template<typename action_t>
template<typename callable_t>
requires std::is_void_v<std::invoke_result_t<std::decay_t<callable_t>&>>
bool untangle::actuator< action_t >::add_task (
                callable_t && callable)   [inline]
```

Adds a nullary callable that returns nothing, as a task that answers true once it has run.

Template Parameters

callable_t	- The callable's type.

Parameters

callable	- The callable to add. An empty one is refused, as by the overload above.

Returns

true - taken, and call_tasks() will fire it.

false - refused: it is empty.

10.1.4.17 remove() [1/2]

```
template<typename action_t>
bool untangle::actuator< action_t >::remove (
                const action_t * action)   [inline]
```

Remove an action from the actions list.

An invalid action (empty std::function) is implicitly removed when operator()() is invoked.

An action the actuator owns is destroyed by this call, so the handle to it is dangling afterwards. See release_owned().

Parameters

action	- Action to be removed. For an owned action this is the handle returned by add(action_t&&).

Returns

true - removed. false - refused during a dispatch (see the re-entrancy convention on actuator), or not in the actions list.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(0);
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.remove(&action1);
actuator_rotate(50);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

10.1.4.18 remove() [2/2]

```
template<typename action_t>
bool untangle::actuator< action_t >::remove (
                const std::string & name)   [inline]
```

Remove an action from actions map.

An action the actuator owns is destroyed by this call, so the handle to it is dangling afterwards. See release_owned().

Parameters

name	- Name of the action to remove.

Returns

true - removed. false - refused during a dispatch (see the re-entrancy convention on actuator), or no action has that name.

10.1.4.19 is_connected()

template<typename action_t>
bool untangle::actuator< action_t >::is_connected () const [inline]

Check if this actuator is ”connected” with other actions.

Remarks

It answers for the actions and for nothing else. Tasks are not actions and do not make an actuator connected; ask has_tasks() about those. Kept this way deliberately when tasks arrived, because a caller that will never hold a task – an attachment, a poll – reads this and must keep reading the answer it always read.

Returns

true - if the actuator::actions list is not empty.

false - if the actuator::actions list is empty.

10.1.4.20 has_tasks()

template<typename action_t>
bool untangle::actuator< action_t >::has_tasks () const [inline]

Is there a task waiting to be fired?

The counterpart to is_connected(), and the two answer different questions: this one is about actuator::tasks, that one about actuator::actions. An actuator can hold either, both, or neither.

Remarks

It is what a queue asks to decide whether its worker still has work. Reading is_connected() for that would report a batch of tasks as nothing to do, and the work would sit unfired.

A refused task is not a waiting one: add_task() answering false means the task was dropped, so this stays false.

Returns

true - a task is waiting; call_tasks() has something to fire.

false - no task is waiting. True again after call_tasks(), which consumes them.

10.1.4.21 has_action()

template<typename action_t>
bool untangle::actuator< action_t >::has_action (
const std::string & name) const [inline]

Check if there is certain named action.

Parameters

name	- Name associated with the action.

Returns

true - if name can be found in actuator::actions_map

false - if name can not be found in actuator::actions_map

10.1.5 Member Data Documentation

10.1.5.1 tasks

template<typename action_t>
tasks_t untangle::actuator< action_t >::tasks

Tasks waiting to be fired, in the order add_task() took them.

Remarks

Held beside actions and sharing nothing with them: operator()() fires the actions and leaves these alone, call_tasks() fires these and leaves the actions alone.

10.1.5.2 owned

template<typename action_t>
std::list<action_t> untangle::actuator< action_t >::owned

Actions this actuator owns, as added by add(action_t&&).

Remarks

It is a std::list because inserting into one never invalidates the address of an element already in it, so every handle already handed out stays valid as further actions are added. A std::vector would reallocate and dangle all of them at once.

10.1.5.3 dispatching

template<typename action_t>
std::size_t untangle::actuator< action_t >::dispatching = 0

How many operator()() or invoke_action() calls are calling an action: more than one when an action dispatches this actuator again. See the re-entrancy convention.

The documentation for this struct was generated from the following file:

- actuator.hpp

10.2 untangle::invalid_action Struct Reference

Invalid action exception.

#include <actuator.hpp>

Public Member Functions

- invalid_action (std::string text)
  Construct a new invalid action object.
- const char * what () const noexcept override
  The message text describing the reason of this exception.

10.2.1 Detailed Description

Invalid action exception.

Remarks

An action may be provided as a binding to a class function member, by using untangle::bind(). When the class object gets invalid, invoking the action will raise an exception to this type.

10.2.2 Constructor & Destructor Documentation

10.2.2.1 invalid_action()

untangle::invalid_action::invalid_action (
std::string text) [inline], [explicit]

Construct a new invalid action object.

Parameters

text	- A message text, describing the reason of this exception.

10.2.3 Member Function Documentation

10.2.3.1 what()

const char * untangle::invalid_action::what () const [inline], [override], [noexcept]

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- actuator.hpp

10.3 untangle::actuator< action_t >::no_callback Struct Reference

#include <actuator.hpp>

10.3.1 Detailed Description

template<typename action_t>
struct untangle::actuator< action_t >::no_callback

What take_callback() holds when there is no callback: invoke_callback() cannot call it, so it does nothing.

The documentation for this struct was generated from the following file:

- actuator.hpp

Chapter 11

File Documentation

11.1 actuator.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #ifndef UNTANGLE_ACTUATOR_ACTUATOR_HPP
00007 #define UNTANGLE_ACTUATOR_ACTUATOR_HPP
00008
00009 #include <cstddef>
00010 #include <deque>
00011 #include <exception>
00012 #include <functional>
00013 #include <list>
00014 #include <map>
00015 #include <memory>
00016 #include <string>
00017 #include <tuple>
00018 #include <type_traits>
00019 #include <utility>
00020 #include <vector>
00021
00022 namespace untangle {
00023 // exception
00032 struct invalid_action : std::exception {
00038   explicit invalid_action(std::string text) : message(std::move(text)) {}
00039
00045   const char* what() const noexcept override { return message.c_str(); }
00046
00047  private:
00048   std::string message;
00049 };
00050
00058 template <typename... Args>
00059 auto last_arg(Args&&... args) {
00060   if constexpr (sizeof...(Args) <= 0) {
00061     return 0;
00062   } else {
00063     return std::get<sizeof...(Args) - 1>(std::forward_as_tuple(args...));
00064   }
00065 }
00066
00085 template <typename callback_t, typename result_t>
00086 concept task_callback_for = requires(callback_t& c, result_t&& r) {
00087   requires std::is_void_v<decltype(c(std::forward<result_t>(r)))>;
00088 } || (std::is_void_v<result_t> && requires(callback_t& c) {
00089                                     requires std::is_void_v<decltype(c())>;
00090                                   });
00091
00107 template <typename callable_t>
00108 concept testable_for_emptiness =
00109     std::is_pointer_v<callable_t> || requires(const callable_t& c) { c.operator bool(); };
00110
00127 using task_t = std::function<bool()>;
00128
00187 template <typename action_t>
00188 struct actuator final {
00200   using actions_t = std::vector<action_t*>;
00201   using actions_map_t = std::map<std::string, action_t*>;
00213   using tasks_t = std::deque<task_t>;
00214   using result_t = std::conditional<std::is_void<typename action_t::result_type>::value, int,
```

```
00215                                 typename action_t::result_type>;
00225   using results_t = std::vector<typename result_t::type>;
00231   using errors_t = std::vector<std::exception_ptr>;
00232
00233   actions_t actions;
00234   actions_map_t actions_map;
00235   results_t results;
00236   errors_t errors;
00237
00244   tasks_t tasks;
00245
00253   std::list<action_t> owned;
00254
00257   std::size_t dispatching = 0;
00258
00259   actuator() = default;
00265   actuator(const actuator& other) { copy_from(other); }
00266   actuator(actuator&&) noexcept = default;
00267   ~actuator() = default;
00274   actuator& operator=(const actuator& other) {
00275     if (this != &other) {
00276       copy_from(other);
00277     }
00278     return *this;
00279   }
00280   actuator& operator=(actuator&&) noexcept = default;
00281
00289   action_t type() const { return nullptr; }
00290
00302   bool reset() {
00303     if (dispatching > 0) {
00304       return false;
00305     }
00306     actions.clear();
00307     actions_map.clear();
00308     tasks.clear();
00309     results.clear();
00310     errors.clear();
00311     owned.clear();
00312     return true;
00313   }
00314
00329   void copy_from(const actuator& other) {
00330     owned = other.owned;
00331
00332     // Old address -> new address, for the owned actions only.
00333     std::map<const action_t*, action_t*> remap;
00334     auto src = other.owned.begin();
00335     auto dst = owned.begin();
00336     for (; src != other.owned.end(); ++src, ++dst) {
00337       remap.emplace(&*src, &*dst);
00338     }
00339     const auto translate = [&remap](action_t* action) {
00340       const auto it = remap.find(action);
00341       return it != remap.end() ? it->second : action;
00342     };
00343
00344     actions.clear();
00345     for (const auto& action : other.actions) {
00346       actions.push_back(translate(action));
00347     }
00348     actions_map.clear();
00349     for (const auto& entry : other.actions_map) {
00350       actions_map.emplace(entry.first, translate(entry.second));
00351     }
00352     // Named explicitly, like every member here: this function does not default, so anything it
00353     // does not copy is silently lost by every copy of an actuator.
00354     tasks = other.tasks;
00355     results = other.results;
00356     errors = other.errors;
00357   }
00358
00373   void release_owned(const action_t* action) {
00374     for (const auto& a : actions) {
00375       if (a == action) {
00376         return;
00377       }
00378     }
00379     for (const auto& entry : actions_map) {
00380       if (entry.second == action) {
00381         return;
00382       }
00383     }
00384     owned.remove_if([action](const action_t& a) { return &a == action; });
00385   }
00386
00442   template <typename last_t>
```

```
00443   void invoke_callback(last_t& last) {
00444     if constexpr (requires { requires std::is_void_v<decltype(last(results.back()))>; }) {
00445       last(results.back());
00446     }
00447   }
00448
00451   struct no_callback {};
00452
00466   template <typename... Args>
00467   static auto take_callback(Args&... args) {
00468     using result_t = typename action_t::result_type;
00469
00470     if constexpr (sizeof...(Args) == 0 || std::is_void_v<result_t>) {
00471       return no_callback{};
00472     } else {
00473       using last_t = std::decay_t<std::tuple_element_t<sizeof...(Args) - 1, std::tuple<Args...»>;
00474
00475       if constexpr (requires(last_t& c, result_t& r) { requires std::is_void_v<decltype(c(r))>; }) {
00476         return last_arg(args...);
00477       } else {
00478         return no_callback{};
00479       }
00480     }
00481   }
00482
00498   template <typename... Args>
00499   void operator()(Args&&... args) {
00500     results.clear();
00501     errors.clear();
00502
00503     // Copied up front, before any action can move the callback out of the argument pack.
00504     auto last = take_callback(args...);
00505
00506     // Dead bindings are collected here and dropped after the loop.
00507     // The actuator does not own the actions it points at, so it must never
00508     // write through action_t* into a std::function belonging to the caller.
00509     std::vector<action_t*> dead_actions;
00510
00511     ++dispatching;
00512     // By index, the size re-read: an action added during the dispatch still runs in this pass.
00513     for (std::size_t i = 0; i < actions.size(); ++i) {
00514       action_t* const action = actions[i];
00515       // A null pointer or an empty std::function can never be invoked. Calling an
00516       // empty one throws std::bad_function_call, which is not an invalid_action and
00517       // would escape this operator, so drop it instead of invoking it.
00518       if (action == nullptr || !*action) {
00519         dead_actions.push_back(action);
00520         continue;
00521       }
00522
00523       try {
00524         if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00525           (*action)(std::forward<Args>(args)...);
00526         } else {
00527           results.push_back((*action)(std::forward<Args>(args)...));
00528           invoke_callback(last);
00529         }
00530       } catch (const invalid_action&) {
00531         // A dead binding is not coming back: it is recorded and the action is dropped.
00532         errors.push_back(std::current_exception());
00533         dead_actions.push_back(action);
00534       } catch (...) {
00535         // Anything else is the action's own failure. It is recorded, the action is kept, and the
00536         // actions behind it still run.
00537         errors.push_back(std::current_exception());
00538       }
00539     }
00540
00541     --dispatching;
00542
00543     // Refused inside a dispatch further out; the next dispatch finds them dead again.
00544     for (const auto& dead_action : dead_actions) {
00545       remove(dead_action);
00546     }
00547   }
00548
00565   template <typename... Args>
00566   void invoke_action(const std::string& name, Args&&... args) {
00567     results.clear();
00568     errors.clear();
00569
00570     // Copied up front, before the action can move the callback out of the argument pack.
00571     auto last = take_callback(args...);
00572     const auto it = actions_map.find(name);
00573     if (it == actions_map.end()) {
00574       return;
00575     }
```

```
00576
00577     // A null pointer or an empty std::function can never be invoked, and dereferencing a
00578     // null one to try is undefined behaviour before a call is even made. Calling an empty
00579     // one throws std::bad_function_call, which is not an invalid_action and would escape
00580     // this method. Drop it instead, as operator()() does for the actions list.
00581     if (it->second == nullptr || !*it->second) {
00582       remove(name);
00583       return;
00584     }
00585
00586     ++dispatching;
00587     bool dead = false;
00588     try {
00589       if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00590         (*it->second)(std::forward<Args>(args)...);
00591       } else {
00592         results.push_back((*it->second)(std::forward<Args>(args)...));
00593         invoke_callback(last);
00594       }
00595     } catch (const invalid_action&) {
00596       errors.push_back(std::current_exception());
00597       dead = true;
00598     } catch (...) {
00599       errors.push_back(std::current_exception());
00600     }
00601     --dispatching;
00602
00603     if (dead) {
00604       remove(name);
00605     }
00606   }
00607
00621   void call_tasks() {
00622     // Taken by swap, so a callback that adds a task adds it to the *next* pass. Iterating the
00623     // member instead would let a task that re-adds itself keep this loop from ever ending -- and
00624     // it is the same reason a queue built on this takes its batch by move.
00625     tasks_t firing;
00626     firing.swap(tasks);
00627
00628     for (auto& task : firing) {
00629       try {
00630         // The task notifies its own callback: bind_task() sealed the two together. What it answers
00631         // is for callers that want it; the actuator does not.
00632         (void)task();
00633       } catch (...) {
00634         // The task's own failure, or its callback's -- the two are not told apart here, and the
00635         // attention above says why.
00636         errors.push_back(std::current_exception());
00637       }
00638     }
00639   }
00640
00649   void add(action_t* action) { actions.push_back(action); }
00650
00666   action_t* add(action_t&& action) {
00667     owned.push_back(std::move(action));
00668     actions.push_back(&owned.back());
00669     return &owned.back();
00670   }
00671
00678   void add(const std::string& name, action_t* action) { actions_map.emplace(name, action); }
00679
00690   action_t* add(const std::string& name, action_t&& action) {
00691     owned.push_back(std::move(action));
00692     const auto [it, inserted] = actions_map.emplace(name, &owned.back());
00693     if (!inserted) {
00694       owned.pop_back();
00695       return nullptr;
00696     }
00697     return &owned.back();
00698   }
00699
00718   bool add_task(task_t&& task) {
00719     if (!task) {
00720       return false;
00721     }
00722
00723     tasks.push_back(std::move(task));
00724     return true;
00725   }
00726
00737   template <typename callable_t>
00738     requires std::is_void_v<std::invoke_result_t<std::decay_t<callable_t>&»
00739   bool add_task(callable_t&& callable) {
00740     if constexpr (testable_for_emptiness<std::decay_t<callable_t>>) {
00741       if (!callable) {
00742         return false;
```

```
00743       }
00744     }
00745
00746     return add_task(task_t([run = std::forward<callable_t>(callable)]() mutable {
00747       run();
00748       return true;
00749     }));
00750   }
00751
00768   bool remove(const action_t* action) {
00769     if (dispatching > 0) {
00770       return false;
00771     }
00772     const auto removed = std::erase(actions, action);
00773     release_owned(action);
00774     return removed > 0;
00775   }
00776
00787   bool remove(const std::string& name) {
00788     if (dispatching > 0) {
00789       return false;
00790     }
00791     const auto it = actions_map.find(name);
00792     if (it == actions_map.end()) {
00793       return false;
00794     }
00795     const action_t* action = it->second;
00796     actions_map.erase(it);
00797     release_owned(action);
00798     return true;
00799   }
00800
00812   bool is_connected() const { return !actions.empty() || !actions_map.empty(); }
00813
00831   bool has_tasks() const { return !tasks.empty(); }
00832
00834   bool is_dispatching() const { return dispatching > 0; }
00835
00843   bool has_action(const std::string& name) const {
00844     return actions_map.find(name) != actions_map.end();
00845   }
00846 };
00847
00865 template <typename action_t, typename arg_t>
00866 void connect_one(actuator<action_t>& target, arg_t&& arg) {
00867   if constexpr (std::is_lvalue_reference_v<arg_t&&> &&
00868                 std::is_same_v<std::decay_t<arg_t>, action_t>) {
00869     if (arg != nullptr) {
00870       target.add(&arg);
00871     }
00872   } else {
00873     action_t action(std::forward<arg_t>(arg));
00874     if (action != nullptr) {
00875       target.add(std::move(action));
00876     }
00877   }
00878 }
00879
00922 template <typename action_t, typename... Actions>
00923 actuator<action_t> connect(action_t& A1, Actions&&... An) {
00924   actuator<action_t> target;
00925   connect_one(target, A1);
00926   (connect_one(target, std::forward<Actions>(An)), ...);
00927   return target;
00928 }
00929
00946 template <typename action_t, typename... Actions>
00947 actuator<action_t> connect(std::type_identity_t<action_t>&& A1, Actions&&... An) {
00948   actuator<action_t> target;
00949   connect_one(target, std::move(A1));
00950   (connect_one(target, std::forward<Actions>(An)), ...);
00951   return target;
00952 }
00953
00954 template <typename actuator_t>
00955 void remove_empty_actions(actuator_t& actuator) {
00956   std::vector<typename actuator_t::actions_map_t::const_iterator> removable_iterators;
00957   for (auto it = actuator.actions_map.begin(); it != actuator.actions_map.end(); ++it) {
00958     if (it->second == nullptr) {
00959       removable_iterators.push_back(it);
00960     }
00961   }
00962   for (auto it : removable_iterators) {
00963     actuator.actions_map.erase(it);
00964   }
00965 }
00966
```

```
00984 template <typename action_t, typename entry_t>
00985 void connect_named_one(actuator<action_t>& target, entry_t&& entry) {
00986   if constexpr (std::is_same_v<std::remove_cvref_t<decltype(entry.second)>, action_t*>) {
00987     if (entry.second != nullptr) {
00988       target.add(entry.first, entry.second);
00989     }
00990   } else {
00991     action_t action(std::forward<entry_t>(entry).second);
00992     if (action != nullptr) {
00993       target.add(entry.first, std::move(action));
00994     }
00995   }
00996 }
00997
01032 template <typename action_t, typename key_t, typename... Actions>
01033 actuator<action_t> connect(std::pair<key_t, action_t*> A1, Actions&&... An) {
01034   actuator<action_t> target;
01035   connect_named_one(target, std::move(A1));
01036   (connect_named_one(target, std::forward<Actions>(An)), ...);
01037   return target;
01038 }
01039
01056 template <typename action_t, typename key_t, typename value_t, typename... Actions>
01057   requires(!std::is_same_v<value_t, action_t*>)
01058 actuator<action_t> connect(std::pair<key_t, value_t> A1, Actions&&... An) {
01059   actuator<action_t> target;
01060   connect_named_one(target, std::move(A1));
01061   (connect_named_one(target, std::forward<Actions>(An)), ...);
01062   return target;
01063 }
01064
01065 // generic helpers to remove the const and noexcept qualifiers from a function type,
01066 // for instance const member functions
01067 template <typename T>
01068 struct function_remove_const;
01069
01070 template <typename R, typename... Args>
01071 struct function_remove_const<R(Args...)> {
01072   using type = R(Args...);
01073 };
01074
01075 template <typename R, typename... Args>
01076 struct function_remove_const<R(Args...) const> {
01077   using type = R(Args...);
01078 };
01079
01080 template <typename R, typename... Args>
01081 struct function_remove_const<R(Args...) noexcept> {
01082   using type = R(Args...);
01083 };
01084
01085 template <typename R, typename... Args>
01086 struct function_remove_const<R(Args...) const noexcept> {
01087   using type = R(Args...);
01088 };
01089
01093
01115 template <typename class_t, typename T,
01116           typename action_t = std::function<typename function_remove_const<T>::type»
01117 action_t bind(const std::shared_ptr<class_t>& obj, T class_t::* method) {
01118   return [wp = std::weak_ptr<class_t>(obj), method](auto&&... args) ->
01119         typename action_t::result_type {
01120           // lock() also keeps the object alive for the duration of the call
01121           const auto obj_ = wp.lock();
01122           if (!obj_) {
01123             // inform the actuator about dead binding
01124             throw invalid_action("bind: invalid object");
01125           }
01126           return ((*obj_).*method)(std::forward<decltype(args)>(args)...);
01127         };
01128 }
01129
01145 template <typename class_t, typename R, typename... Args>
01146 std::function<R(Args...)> bind(const std::shared_ptr<class_t>& obj,
01147                                std::function<R(Args...)> class_t::* field) {
01148   return [wp = std::weak_ptr<class_t>(obj), field](auto&&... args) -> R {
01149     // lock() also keeps the object alive for the duration of the call
01150     const auto obj_ = wp.lock();
01151     if (!obj_) {
01152       throw invalid_action("bind: invalid object");
01153     }
01154     const auto& function = (*obj_).*field;
01155     if (!function) {
01156       throw invalid_action("bind: empty field");
01157     }
01158     return function(std::forward<decltype(args)>(args)...);
01159   };
```

```
01160 }
01161
01182 template <typename class_t, typename T,
01183           typename action_t = std::function<typename function_remove_const<T>::type»
01184 action_t bind(class_t* obj, T class_t::* method) {
01185   return [obj, method](auto&&... args) -> typename action_t::result_type {
01186     if (!obj) {
01187       // inform the actuator about dead binding
01188       throw invalid_action("bind: invalid object");
01189     }
01190     return ((obj)->*method)(std::forward<decltype(args)>(args)...);
01191   };
01192 }
01193
01226 template <typename action_t, typename... Args>
01227 task_t bind_task(action_t&& action, Args&&... args) {
01228   using result_t = typename std::decay_t<action_t>::result_type;
01229
01230   static_assert(sizeof...(Args) > 0,
01231                 "bind_task: a task must be given a callback as its last argument");
01232
01233   constexpr std::size_t last = sizeof...(Args) - 1;
01234   using callback_t = std::tuple_element_t<last, std::tuple<std::decay_t<Args>...»;
01235
01236   static_assert(task_callback_for<callback_t, result_t>,
01237                 "bind_task: the last argument must be a callback taking the task's result and "
01238                 "returning nothing (void() for a task that returns nothing)");
01239
01240   // References into the caller's arguments. What is copied out of it is the task's own, below.
01241   auto pack = std::forward_as_tuple(std::forward<Args>(args)...);
01242
01243   // A task that could not do its job is built empty, and add_task() refuses it on the caller's
01244   // own stack. Only a callable that can answer is asked -- see \ref testable_for_emptiness -- and
01245   // the answer is read contextually, so no explicit conversion is left to truncate.
01246   if constexpr (testable_for_emptiness<callback_t>) {
01247     if (!std::get<last>(pack)) {
01248       return {};
01249     }
01250   }
01251   if constexpr (testable_for_emptiness<std::decay_t<action_t>>) {
01252     if (!action) {
01253       return {};
01254     }
01255   }
01256
01257   // The pack is indexed rather than unpacked, because everything but its last element is bound and
01258   // a fold cannot say "all but the last". The bound arguments are copied into one tuple rather
01259   // than into a capture pack: an init-capture pack expanded in a dependent context crashes GCC 14
01260   // (an ICE in tsubst_pack_expansion), and a tuple says the same thing in an ordinary expression.
01261   return [&]<std::size_t... i>(std::index_sequence<i...>) -> task_t {
01262     return [action = std::forward<action_t>(action), callback = callback_t(std::get<last>(pack)),
01263             bound = std::tuple<std::decay_t<decltype(std::get<i>(pack))>...>(
01264                 std::get<i>(pack)...)]() mutable -> bool {
01265       if constexpr (std::is_void_v<result_t>) {
01266         std::apply(action, bound);
01267         callback();
01268       } else {
01269         // One expression, so the result is handed over as an rvalue: a result type that cannot be
01270         // copied is still a result a task can report.
01271         callback(std::apply(action, bound));
01272       }
01273       return true;
01274     };
01275   }(std::make_index_sequence<last>{});
01276 }
01277
01278 }  // namespace untangle
01279
01280 #endif  // UNTANGLE_ACTUATOR_ACTUATOR_HPP
```

Index

actions_t
    untangle::actuator< action_t >, 34
actuator, 1
    untangle::actuator< action_t >, 35
add
    untangle::actuator< action_t >, 40, 41, 42
add_task
    untangle::actuator< action_t >, 42, 43

bind
    namespace untangle: functions, 19, 20
bind_task
    untangle, 26

call_tasks
    untangle::actuator< action_t >, 40
connect
    namespace untangle: functions, 16, 17, 18
connect_named_one
    untangle, 26
connect_one
    untangle, 25
copy_from
    untangle::actuator< action_t >, 36

dispatching
    untangle::actuator< action_t >, 46

errors_t
    untangle::actuator< action_t >, 35

has_action
    untangle::actuator< action_t >, 45
has_tasks
    untangle::actuator< action_t >, 45

invalid_action
    untangle::invalid_action, 47
invoke_action
    untangle::actuator< action_t >, 39
invoke_callback
    untangle::actuator< action_t >, 37
is_connected
    untangle::actuator< action_t >, 44

last_arg
    untangle, 25

namespace untangle: functions, 15
    bind, 19, 20
    connect, 16, 17, 18

operator()
    untangle::actuator< action_t >, 39
operator=
    untangle::actuator< action_t >, 36
owned
    untangle::actuator< action_t >, 46

release_owned
    untangle::actuator< action_t >, 37
remove
    untangle::actuator< action_t >, 43, 44
reset
    untangle::actuator< action_t >, 36
results_t
    untangle::actuator< action_t >, 35

take_callback
    untangle::actuator< action_t >, 39
task_t
    untangle, 24
tasks
    untangle::actuator< action_t >, 46
tasks_t
    untangle::actuator< action_t >, 34
type
    untangle::actuator< action_t >, 36

untangle, 23
    bind_task, 26
    connect_named_one, 26
    connect_one, 25
    last_arg, 25
    task_t, 24
untangle::actuator< action_t >, 31
    actions_t, 34
    actuator, 35
    add, 40, 41, 42
    add_task, 42, 43
    call_tasks, 40
    copy_from, 36
    dispatching, 46
    errors_t, 35
    has_action, 45
    has_tasks, 45
    invoke_action, 39
    invoke_callback, 37
    is_connected, 44
    operator(), 39
    operator=, 36
    owned, 46

release_owned, 37
remove, 43, 44
reset, 36
results_t, 35
take_callback, 39
tasks, 46
tasks_t, 34
type, 36
untangle::actuator< action_t >::no_callback, 48
untangle::invalid_action, 47
invalid_action, 47
what, 47
untangle::task_callback_for, 29
untangle::testable_for_emptiness, 30

what
untangle::invalid_action, 47